

Weekly Notice 9

Exercises, week 9:

1. Let U and V denote independent and identically distributed real-valued square-integrable random variables. Which of the following statements are necessarily true? If a statement is not true, give a counterexample. If it is true, give a brief argument.
 - (a) $U + V \stackrel{d}{=} 2U$.
 - (b) $\mathbb{E}[U + V] = 2\mathbb{E}[U]$.
 - (c) $\text{Var}(U + V) = 2 \text{Var}(U)$.
 - (d) $\text{Var}(U + V) = 4 \text{Var}(U)$.
2. Exercise 92. (Be careful – many students guess incorrectly. You may want to look at the previous exercise.)
3. Exercise 96.
4. Exercise 91(2)–(3).
5. Exercise 94.
6. Exercise 99–100.

Remark: Under the assumptions in Exercise 96(1), we actually have $E[Y_t] = \mu t$ for every $t \geq 0$.